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Machine Learning for Out-of-Sample Return Forecasting and Portfolio Allocation

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Abstract

This thesis examines whether machine learning and regularized linear models improve monthly out-of-sample return forecasting and generate economically valuable portfolio strategies for U.S. industry portfolios. The analysis compares the historical mean benchmark, unrestricted and restricted predictive regressions, Ridge regression, Elastic Net, and Random Forest using an expanding-window forecasting framework, with a rolling-window specification used as a robustness check. Forecast performance is evaluated using mean squared prediction error, out-of-sample R^2 , Clark-West tests against the historical mean benchmark, and Diebold-Mariano tests for selected model comparisons. Economic value is assessed by constructing mean-variance portfolios under long-only constraints, weight caps, shrinkage toward the $1/N$ benchmark, and transaction costs. The results show that monthly return forecasting remains difficult: unrestricted OLS performs poorly, non-negativity restrictions improve linear forecasts, and Ridge provides the strongest and most robust forecast performance. Random Forest does not consistently outperform traditional or regularized linear models. Portfolio results are more favorable, especially for capped OLS, Random Forest, and Elastic Net strategies before transaction costs. However, under high transaction costs, the $1/N$ benchmark remains the most robust strategy.

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1 Introduction

Predicting stock returns is one of the central challenges in empirical finance. Reliable forecasts could improve portfolio allocation, strengthen risk management, and help investors respond more effectively to changing market conditions. Although many macro-financial variables have been proposed as predictors of future returns, return prediction remains difficult. Traditional forecasting models such as Ordinary Least Squares (OLS) often produce unstable results when evaluated out-of-sample, while simple benchmarks such as the historical mean remain hard to beat (Welch and Goyal, 2008). This creates a clear practical challenge: can more advanced forecasting methods extract useful predictive information without being overwhelmed by the noise in financial returns?

This thesis examines two approaches that may offer a solution. Regularized linear models, such as Ridge regression and Elastic Net, aim to improve stability by shrinking coefficient estimates, while machine learning models, such as Random Forests, can capture nonlinear relationships and interactions that traditional regressions may miss (Gu et al., 2020; ?). The analysis evaluates whether these methods improve monthly out-of-sample forecasts of U.S. industry excess returns relative to traditional benchmarks and whether any forecasting gains translate into economically meaningful portfolio performance after accounting for risk, turnover, and transaction costs. However, the potential of these methods must be considered against practical difficulties that historically have limited return prediction. Traditional predictive regressions, while valued for their interpretability, often suffer from structural instability and estimation risk, causing them to perform poorly out-of-sample (Welch and Goyal, 2008; Rapach et al., 2010). Because return variation is largely driven by unpredictable news, parameter-heavy models may overfit noisy signals rather than capture persistent risk premia (Gu et al., 2020). Consequently, the historical mean forecast and the $1/N$ rule remain formidable benchmarks because they require little or no parameter estimation; therefore, avoid much of the estimation uncertainty that destabilizes more sophisticated models (DeMiguel et al., 2009; Tu and Zhou, 2011).

Restrictions and regularization methods such as Ridge regression and Elastic Net can improve linear models by shrinking coefficients, reducing overparameterization, and selecting more stable predictors (Gu et al., 2020). However, these models may still struggle when

predictive relationships are nonlinear or depend on interactions among variables. Machine learning methods offer greater flexibility in capturing such patterns, yet this same flexibility also increases the risk of fitting noise rather than persistent information (Gu et al., 2020; ?).

Crucially, the accuracy of the statistical forecast does not always translate into superior portfolio results. A model may produce only limited improvements in out-of-sample R^2 while still supporting useful portfolio decisions, or it may forecast well statistically but lose its economic value through unstable weights, high turnover, and transaction costs (Campbell and Thompson, 2008). A rigorous comparison is therefore necessary to determine whether modern forecasting methods provide an advantage to investors.

Against this background, the objective of this thesis is to evaluate whether more advanced forecasting methods can improve both the statistical accuracy and the economic usefulness of monthly return predictions. In particular, the study compares regularized linear models and machine learning methods with traditional forecasting benchmarks to determine whether they produce more accurate out-of-sample forecasts and whether those forecasts can support economically valuable portfolio decisions. Therefore, the central research question is:

Can machine learning and regularized linear models outperform traditional benchmarks in monthly out-of-sample return forecasting, and can their forecasts generate economically valuable portfolio strategies?

To answer this question, the study considers the following sub-questions:

1. Do non-negativity restrictions improve the out-of-sample performance of linear forecasting models?
2. Do regularized linear models, particularly Ridge regression, outperform OLS and the historical mean benchmark?
3. Does Random Forest outperform the historical mean and the strongest linear forecasting models?
4. Do improvements in forecast accuracy translate into better portfolio performance?

5. Do any portfolio gains remain after accounting for turnover and transaction costs?

The remainder of this thesis is organized as follows. The next section reviews the relevant literature. The methodology section then describes the data, forecasting models, portfolio construction, and evaluation measures. The results section presents the out-of-sample forecasting and portfolio performance findings, followed by robustness checks. Finally, the conclusion interprets the results in relation to the existing literature and summarizes the main findings and implications.

2 Literature review

2.1 Traditional Return Prediction Models

The prediction of stock returns has been a central topic in financial research for over a century, with early work already examining the systematic role of valuation ratios (Dow, 1920). This line of inquiry was later formalized in the late 1980s, when researchers showed that interest-rate variables and yield spreads could be used in linear frameworks to track time-varying expected returns associated with the business cycle (Campbell, 1987; Fama and French, 1989).

Traditional methodologies have relied heavily on Ordinary Least Squares (OLS) predictive regressions because their theoretical simplicity and the direct interpretability of their coefficients allow for clear economic analysis (Awijen et al., 2023). However these linear models often show statistical significance within a training sample, they frequently fail to provide reliable out-of-sample forecasts (Wang and Zhou, 2023). Welch and Goyal (2008) demonstrate that many predictive regressions underperform the simple historical average benchmark over long out-of-sample periods, suggesting that apparent predictive relationships may be spurious, unstable, or sample-specific.

A primary weakness of the OLS framework is the assumption of a stable, time-invariant relationship between predictors and future returns. Recent research by Yu and Yan (2025) provides evidence that predictive coefficients fluctuate over time, as return predictability is often episodic and dependent on the prevailing economic regime. Consequently, standard linear models may struggle to adapt to structural shifts in the economy, contributing to

unstable out-of-sample performance across different market environments (Hollstein et al., 2025). Furthermore, as the number of potential predictors has increased, the traditional “kitchen sink” OLS approach has become increasingly prone to estimation risk and overfitting. Babiak and Baruník (2026) argue that strong statistical performance within a sample may vanish during real-time implementation because OLS may fit idiosyncratic noise rather than persistent signals.

2.2 Restricted Predictive Regressions

Rather than abandoning predictive regressions, subsequent research has examined whether their weaknesses can be reduced through economically motivated constraints. Campbell and Thompson (2008) introduce weak restrictions based on investment theory, including setting coefficients with theoretically incorrect signs equal to zero and truncating negative equity-premium forecasts at zero. These restrictions are designed to rule out economically implausible predictions and improve out-of-sample performance when the reduction in estimation variance outweighs the bias introduced by the constraint.

The main value of restricted predictive regressions is that they preserve the transparency of linear models while reducing the effect of unstable or noisy forecasts. In an equity-premium forecasting setting, Campbell and Thompson (2008) show that theoretically motivated restrictions help standard predictors outperform the historical mean benchmark. Similarly, Tsiakas et al. (2020) and Yuan et al. (2025), adds that theory and economically based constraints can make forecasts more stable and uncover predictive information from fundamental variables across different market environments. Recent extensions also suggest that tighter or more forward-looking restrictions can improve forecast stability by filtering out economically unreasonable signals during turbulent market conditions (Wang and Zhou, 2023).

2.3 Regularized Linear Models

Regularized linear models provide an intermediate approach between unrestricted OLS regressions and more complex nonlinear machine learning methods. By using shrinkage and variable selection, they can improve forecast stability when predictors are numerous, cor-

related, persistent, or weak. In a closely related equity-premium forecasting setting, [Li and Tsiakas \(2017\)](#) show that combining statistical shrinkage with economic restrictions improves out-of-sample predictability and generates economically meaningful certainty-equivalent gains. Similarly, [Lee et al. \(2022\)](#) show that accounting for differences in predictor persistence improves variable selection and leads to more reliable out-of-sample performance. More recent work reaches similar conclusions: [Maung and Swanson \(2025\)](#) emphasize the value of shrinkage when predictive regressions contain many uncertain signals, while [Gao et al. \(2026\)](#) show that regularized methods are especially useful in high-dimensional predictive-regression settings.

Broader return-prediction research also supports the use of regularization as a stronger alternative to unrestricted OLS. [Yuan et al. \(2025\)](#) prove that penalized linear models, including Ridge and Elastic Net, can improve out-of-sample forecasting performance by reducing overfitting and model uncertainty across changing market conditions. Similarly, [Awijen et al. \(2023\)](#) recognize that post-selection linear models can deliver economically meaningful return predictability by focusing on relevant predictors and reducing noise. Together, these findings suggest that Ridge and Elastic Net provide demanding linear benchmarks for assessing whether Random Forest adds value through nonlinear relationships and interactions that cannot be captured by a well-regularized regression model.

2.4 Machine Learning in Asset Pricing

Machine learning has become increasingly important in empirical asset pricing because expected-return estimation is fundamentally a prediction problem. [Gu et al. \(2020\)](#) argue that asset pricing is a natural setting for prediction-focused methods, especially when the number of potential signals is large and relationships are difficult to capture with unrestricted linear models. Unlike traditional regressions, machine learning methods can capture nonlinearities and interaction effects without requiring a specific functional form in advance. Recent research continues to support this view: [Babiak and Baruník \(2026\)](#) show that nonlinear models can improve portfolio construction when return dynamics are complex, while [Wang \(2025\)](#) finds that more flexible architectures can outperform simpler neural-network models in return-forecasting applications.

The potential advantage of machine learning is closely related to the idea that model complexity can be valuable when it is properly controlled. [Kelly et al. \(2024\)](#) argue that greater complexity can improve expected out-of-sample performance because richer models are better able to approximate the unknown return-prediction function. Related evidence from [Maung and Swanson \(2025\)](#) suggests that flexible forecasting frameworks are particularly useful when predictive signals are numerous and uncertain, while [Babiarz and Barunik \(2026\)](#) show that highly parameterized models can extract information from complex return dynamics that simpler specifications may overlook. However, these advantages depend on appropriate regularization, since financial returns are noisy and flexible models can easily overfit. This point is especially relevant for this thesis, where Random Forest is compared not only with unrestricted OLS, but also with restricted and regularized linear models.

Random Forests are particularly relevant because they provide a flexible, non-parametric approach to return forecasting while reducing the instability of individual decision trees ([Gu et al., 2020](#)). By averaging predictions across many trees, Random Forests preserve the ability to capture nonlinearities and interactions while limiting variance ([Pinelis and Ruppert, 2022](#)). [Huang et al. \(2026\)](#) find that Random Forest models can outperform robust linear regressions and some deep-learning benchmarks in cross-sectional stock-return prediction, while [Pinelis and Ruppert \(2022\)](#) show that Random Forest-based return and volatility timing can improve risk-adjusted portfolio performance. These findings support the use of Random Forest as a practical machine-learning benchmark for testing whether nonlinear flexibility adds value beyond well-regularized linear models.

2.5 Equal-Weighting and Simple Portfolio Rules

Since Markowitz’s modern portfolio theory, asset allocation has traditionally been framed as an optimization problem in which investors choose portfolio weights by balancing expected returns against risk ([Markowitz, 1952](#)). In practice, however, mean-variance optimization is highly sensitive to estimation error, since small errors in expected returns and covariance estimates can generate extreme and unstable portfolio weights ([DeMiguel et al., 2009](#)). This problem has made the naive diversification rule, or $1/N$ strategy, a robust alternative because it allocates an equal fraction of wealth to each available asset without relying on

estimated return moments (DeMiguel et al., 2009; Tu and Zhou, 2011). DeMiguel et al. (2009) show that a wide range of sophisticated asset allocation models fail to consistently outperform the $1/N$ rule in terms of Sharpe ratio, certainty-equivalent return, and turnover. They further show that, for an optimized portfolio of 25 assets to outperform the $1/N$ benchmark, an estimation window of approximately 3,000 months would be required, which is rarely available in practice. This illustrates how estimation error can overwhelm the theoretical benefits of optimization and makes the $1/N$ rule an important benchmark.

Beyond forecast accuracy, portfolio implementation is also affected by turnover and transaction costs, which can reduce net returns relative to the more passive rebalancing of the $1/N$ rule. Tu and Zhou (2011) propose combining $1/N$ with Markowitz-type and shrinkage-based strategies, treating equal weighting as a shrinkage target that balances the low estimation variance of naive diversification against the potential gains from using return and risk estimates. More recent work also emphasizes that statistical forecast accuracy and portfolio value do not always move together. ? find that a model may still support valuable market-timing decisions even when its out-of-sample R^2 is weak or negative, because portfolio performance depends on how forecasts translate into allocation choices. This highlights why forecast-based strategies must be evaluated through portfolio performance rather than prediction accuracy alone.

3 Methodology

3.1 Empirical Forecasting Framework

The empirical analysis follows a monthly out-of-sample forecasting framework in which information available at time t is used to predict excess returns in month $t + 1$. The main specification uses an expanding estimation window, so each forecast is based on all historical observations available up to that point. This approach is well suited to return prediction because monthly excess returns are noisy and short estimation samples can produce unstable coefficient estimates. It also reflects the real-time forecasting logic commonly used in the return-prediction literature, including Welch and Goyal (2008), Campbell and Thompson (2008), Rapach et al. (2010), and Li and Tsiakas (2017). The resulting forecasts are

evaluated both statistically, through out-of-sample prediction accuracy, and economically, through the performance of portfolio strategies formed from those forecasts. To assess whether the results are sensitive to evolving market conditions, a rolling-window design is later considered as a robustness check.

3.2 Data and Variable Construction

The analysis uses monthly data combining the 10 Industry Portfolios from the Kenneth R. French Data Library with macro-financial predictors from the Goyal–Welch dataset, producing a merged sample from January 1927 to December 2024. The industry portfolios cover Consumer Non-Durables, Consumer Durables, Manufacturing, Energy, High Technology, Telecommunications, Shops, Healthcare, Utilities, and Other. The original predictor set includes the market index level, trailing twelve-month dividends and earnings, book-to-market ratio, short-term Treasury bill rate, long-term government bond yield, corporate bond yields, inflation, long-term bond returns, corporate bond returns, stock variance, net equity issuance, and the risk-free rate.

From these raw variables, several standard return-prediction measures were constructed following the definitions commonly used in the equity-premium prediction literature (Welch and Goyal, 2008; Campbell and Thompson, 2008). The dividend-price ratio is defined as the logarithm of trailing twelve-month dividends divided by the market index, while the earnings-price ratio is defined analogously using earnings. The term spread is measured as the difference between the long-term government bond yield and the Treasury bill rate. The default yield spread is measured as the difference between the yields of the BAA and AAA corporate bonds, while the default return spread is measured as the difference between the returns of corporate bonds and the returns of long-term government bonds. The industry portfolio returns were converted from percentages to decimals, and monthly excess returns were calculated for each portfolio by subtracting the risk-free rate. The final dataset was then aligned by month and cleaned to retain a consistent set of observations for the forecasting analysis.

3.3 Forecasting Models

The forecasting analysis compares several models for predicting monthly excess returns for each of the ten industry portfolios. The models range from a simple historical mean benchmark to nonlinear machine learning approach. For the predictive models, both unrestricted and restricted versions are considered. The restricted versions follow the economic logic of [Campbell and Thompson \(2008\)](#): after a model produces its forecast, any negative predicted excess return is set equal to zero. For our analysis we use six following models:

- **Historical mean benchmark.** The historical mean forecast estimates the expected excess return for month $t + 1$ as the average realized excess return observed up to time t .
- **OLS predictive regression.** Traditional OLS predictive regression that uses the full set of available lagged macro-financial predictors to forecast next-month excess returns.
- **OLS-4 predictive regression.** Traditional OLS regression with four selected predictors: the dividend-price ratio, earnings-price ratio, term spread, and default yield spread. These predictors were selected as foundational benchmarks because they represent prominent valuation and macro-financial signals in the return-predictability literature, serving as standard economic fundamentals established by [Welch and Goyal \(2008\)](#) and theoretically motivated by the frameworks of [Campbell and Thompson \(2008\)](#) and [Tsiakas et al. \(2020\)](#). The OLS-4 model is included as a parsimonious benchmark that retains the main economic channels of the broader predictor set while reducing the overfitting risk of the full OLS model. It does not include size, book-to-market, or momentum because the empirical framework is based on aggregate macro-financial predictors and industry portfolio returns rather than firm stock characteristics.
- **Ridge predictive regression.** Ridge regression is used as a regularized version of the linear predictive model. Instead of minimizing only the sum of squared forecast errors, Ridge adds an L_2 penalty on the size of the coefficients:

$$\min_{\alpha, \beta} \sum_{t=1}^T (r_{i,t+1}^e - \alpha_i - X_t' \beta_i)^2 + \lambda \sum_{j=1}^p \beta_{j,i}^2. \quad (1)$$

Here, X_t denotes the vector of lagged predictors, β_i is the coefficient vector for industry i , and λ controls the strength of the penalty. A larger λ shrinks the coefficients more strongly toward zero, reducing estimation variance but introducing some bias. The regularization strength is selected from a wide grid of candidate values ranging from 10^{-4} to 10^6 . The wide grid allows the model to choose between nearly unrestricted estimates and heavily regularized forecasts, depending on the amount of predictive signal available in each estimation window. Similar searches over shrinkage parameters are common in regularized return-prediction applications (Zou and Hastie, 2005; Gu et al., 2020; ?).

- **Elastic Net predictive regression.** Elastic Net extends regularized linear regression by combining the Ridge L_2 penalty with the Lasso L_1 penalty:

$$\min_{\alpha, \beta} \sum_{t=1}^T (r_{i,t+1}^e - \alpha_i - X_t' \beta_i)^2 + \lambda \left[(1 - \alpha_{EN}) \sum_{j=1}^p \beta_{j,i}^2 + \alpha_{EN} \sum_{j=1}^p |\beta_{j,i}| \right]. \quad (2)$$

The L_2 part shrinks coefficients toward zero, while the L_1 part can set some coefficients exactly equal to zero, creating variable selection. The parameter λ determines the overall strength of regularization, while α_{EN} controls the balance between Ridge-type shrinkage and Lasso-type selection. The penalty strength is selected from 25 logarithmically spaced values between 10^{-8} and 10^{-3} , while the mixing parameter is selected from $\{0.05, 0.10, 0.25, 0.50\}$. The relatively low values of the mixing parameter give greater weight to Ridge-type shrinkage, which helps stabilize forecasts when predictors are correlated, while still allowing some variable selection through the Lasso component (Zou and Hastie, 2005; Gu et al., 2020).

- **Random Forest regression.** Random Forest averages predictions across many decision trees. To control model complexity, the hyperparameters are selected from the following grid: the number of trees is set to 300 or 500, maximum tree depth to 3 or 5, the minimum number of observations in terminal nodes to 5, 10, or 20, the

minimum number of observations required to split a node to 10 or 20, and the number of predictors considered at each split to either 30% of all predictors or the square root of the predictor count. This grid intentionally focuses on relatively shallow trees, since return prediction has a low signal-to-noise ratio and overly deep trees may fit noise rather than persistent predictive relationships (Gu et al., 2020). Due to very large computational time grid is kept small size.

To keep the forecasting exercise realistic while avoiding excessive computational burden, hyperparameters for Ridge, Elastic Net, and Random Forest are re-selected once every five years using time-series validation within the expanding estimation window. All models are re-estimated each month as new data become available.

3.4 Portfolio Construction

To evaluate whether better return forecasts translate into meaningful investment outcomes, the predicted excess returns are converted into monthly portfolio strategies using mean-variance allocation. At each month, every forecasting model produces a vector of expected excess returns for the ten industry portfolios. This forecast vector is combined with a historical covariance matrix estimated from past industry returns available at that date, and portfolio weights are then computed through mean-variance optimization. The resulting portfolios are rebalanced monthly as new forecasts and covariance estimates become available.

The analysis considers several portfolio specifications in order to evaluate both the unconstrained and more implementable versions of the forecast-based strategies. The first specification allows short selling and imposes no additional weight restrictions beyond the requirement that portfolio weights sum to one. This unconstrained version shows the direct outcome of the mean-variance allocation rule. The second specification imposes a long-only constraint, requiring all portfolio weights to be non-negative. The third specification applies an upper weight cap, limiting the maximum allocation to any single industry in order to reduce extreme positions. In addition, the analysis considers portfolio combinations that mix forecast-based mean-variance weights with the naive $1/N$ allocation rule. These mixed portfolios allow the strategy to retain information from the forecasting model while

shrinking the allocation toward a simple and stable benchmark.

Alongside these optimized strategies, the analysis includes the $1/N$ benchmark, which allocates an equal 10 percent share of wealth to each of the ten industry portfolios and therefore requires no return forecasts or covariance estimation. This benchmark provides a simple comparison for assessing whether model-based allocation adds value. To evaluate the practical relevance of the results, portfolio performance is also examined after incorporating transaction costs. This allows the analysis to assess whether any gains remain after accounting for the trading required to rebalance portfolios over time.

3.5 Evaluation Metrics

Model performance is assessed from both a statistical forecasting perspective and an economic portfolio perspective. Forecast accuracy is evaluated using the out-of-sample R^2 and the mean squared prediction error (MSPE). The out-of-sample R^2 measures the proportional reduction in mean squared prediction error relative to the historical mean benchmark:

$$R_{OOS}^2 = 1 - \frac{\sum_{t=T_0}^{T-1} (r_{i,t+1}^e - \hat{r}_{i,t+1}^e)^2}{\sum_{t=T_0}^{T-1} (r_{i,t+1}^e - \bar{r}_{i,t+1}^e)^2}, \quad (3)$$

where $r_{i,t+1}^e$ is the realized excess return of industry i , $\hat{r}_{i,t+1}^e$ is the model forecast, and $\bar{r}_{i,t+1}^e$ is the historical mean forecast based only on information available up to time t . A positive value indicates that the model forecasts more accurately than the historical mean benchmark, while a negative value indicates weaker forecasting performance. MSPE captures the average squared difference between predicted and realized excess returns. In addition, formal forecast comparison tests are used to assess whether differences in predictive performance are statistically meaningful. The Clark-West test is used when comparing predictive regressions against the historical mean benchmark. The test is implemented as a one-sided test, where the alternative hypothesis is that the forecasting model improves accuracy relative to the historical mean. The Diebold-Mariano test is used for pairwise comparisons between competing non-nested forecasting models, such as restricted versus unrestricted models, Ridge versus OLS, or Random Forest versus regularized linear models. In this case, the test compares the squared forecast errors of two models. The main reported results use two-sided p -values to test whether the two models differ significantly in forecast accuracy,

while the sign of the test statistic indicates which model has lower average forecast loss.

Portfolio performance is evaluated using average return, volatility, Sharpe ratio, certainty-equivalent return, and turnover. Average return measures the overall profitability of each strategy, while volatility captures the variability and therefore the risk of portfolio returns. The Sharpe ratio summarizes risk-adjusted performance by relating excess return to return volatility.

$$SR_p = \frac{\bar{r}_p^e}{\sigma_p}, \quad (4)$$

where $\bar{r}_p^e$ is the average excess return of portfolio p , and σ_p is the standard deviation of portfolio excess returns. The certainty-equivalent return translates portfolio performance into the guaranteed return that a risk-averse investor would regard as equally desirable, allowing the analysis to assess whether higher returns compensate sufficiently for additional risk.

$$CER_p = \bar{r}_p^e - \frac{\gamma}{2} \sigma_p^2, \quad (5)$$

where $\bar{r}_p^e$ is the average portfolio excess return, σ_p^2 is the variance of portfolio excess returns, and γ is the investor's coefficient of relative risk aversion. Following common practice in the return-prediction and portfolio-choice literature, γ is set to 3 (Campbell and Thompson, 2008; Rapach et al., 2010; Tu and Zhou, 2011). Finally, turnover measures how strongly portfolio weights change from month to month and is important for understanding the trading intensity of each strategy, especially when transaction costs are incorporated. Monthly turnover is calculated as

$$\text{Turnover}_t = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - \tilde{w}_{i,t}|,$$

where $w_{i,t}$ is the target portfolio weight after rebalancing and $\tilde{w}_{i,t}$ is the drifted weight before rebalancing, defined as

$$\tilde{w}_{i,t} = \frac{w_{i,t-1}(1 + r_{i,t})}{1 + r_{p,t}}.$$

Here, $r_{i,t}$ is the realized return of asset i , $r_{p,t}$ is the portfolio return before rebalancing, and the factor $\frac{1}{2}$ avoids double-counting trades.

These evaluation measures are used to test the hypotheses developed in the literature review by comparing the forecasting accuracy and portfolio performance.

3.6 Robustness Check

To assess whether the main results depend on the expanding-window estimation design, the analysis also considers a 20-year rolling-window specification. Unlike the expanding window, which uses all available historical observations up to time (t), the rolling window uses only the most recent 20 years of monthly data. This allows the forecasting models to adapt more quickly to possible changes in return-predictability relationships.

The robustness check focuses on three questions. First, do the forecasting results remain broadly similar when models are estimated using only recent data? Second, are the differences between rolling- and expanding-window forecasts statistically meaningful? Third, do the main portfolio conclusions remain economically relevant under the alternative estimation window? To answer these questions, the main models are re-estimated using the rolling window and compared with the expanding-window forecasts on the common out-of-sample evaluation period. Diebold–Mariano tests are used to compare the two estimation schemes directly. Finally, the portfolio specifications that performed best in the main analysis are applied to the rolling-window forecasts and compared with the corresponding expanding-window results and the $(1/N)$ benchmark. This keeps the robustness check focused on the sensitivity of the main conclusions rather than selecting new optimal portfolio rules from the rolling-window results.

4 Results

This section presents the empirical results of the out-of-sample forecasting and portfolio allocation analysis. The results are first evaluated in terms of statistical forecast accuracy using MSPE and out-of-sample R^2 . The analysis then examines whether the forecasts translate into economically meaningful portfolio performance. Finally, rolling-window results are reported as a robustness check to assess whether the main findings are sensitive to the expanding-window estimation design.

4.1 Forecast Evaluation

Table 1: Expanding-Window Forecasting Performance Across Industry Portfolios

Model	Restriction	Rel. MSPE	Average R_{OS}^2	Positive R_{OS}^2
OLS	No	1.000	-0.0533	0/10
OLS	Yes	0.965	-0.0155	0/10
OLS-4	No	0.958	-0.0093	1/10
OLS-4	Yes	0.952	-0.0023	2/10
Ridge	No	0.950	0.0015	7/10
Ridge	Yes	0.949	0.0021	7/10
Elastic Net	No	0.973	-0.0221	0/10
Elastic Net	Yes	0.958	-0.0068	2/10
Random Forest	No	0.955	-0.0039	3/10

Note. The table reports average out-of-sample forecasting performance across the ten industry portfolios. Relative MSPE is normalized by the unrestricted OLS model (whose average MSPE is 0.003165), so values below 1 indicate lower average forecast errors than unrestricted OLS. R_{OOS}^2 is measured relative to the historical mean benchmark. The final column reports the number of industries for which the model achieves a positive R_{OOS}^2 . The restriction refers to truncating negative predicted excess-return forecasts at zero. The best-performing model is highlighted in bold.

The expanding-window forecast results show that most models struggle to outperform the historical mean benchmark. Full OLS performs worst, with a negative average R_{OOS}^2 and no industries achieving positive out-of-sample predictive performance. Applying the non-negativity restriction improves all linear models, although the improvement is not sufficient for OLS, OLS-4, or Elastic Net to outperform the historical mean on average. Ridge is the strongest forecasting model, producing the lowest average MSPE, a positive average R_{OOS}^2 , and positive R_{OOS}^2 values in seven out of ten industries. This is consistent with [Welch and Goyal \(2008\)](#) and [Campbell and Thompson \(2008\)](#), who show that traditional predictive regressions often struggle out of sample, and with [Gu et al. \(2020\)](#), who emphasize the importance of regularization in noisy return-prediction settings. In contrast, Random Forest

does not outperform the historical mean benchmark in this setting, with a negative average R^2_{OOS} and positive performance in only three industries. Overall, the results suggest that regularization provides more reliable forecasting gains than the nonlinear Random Forest model in the monthly industry-return forecasting framework.

Table 2: Summary of Clark–West Test Results

Model	Spec.	Rel. Adj. Diff.	Avg. CW Stat.	Median p	Significance at 5%;10%
OLS	Unrestricted	1.000	0.780	0.249	0/10; 2/10
OLS	Restricted	0.580	0.898	0.182	1/10; 1/10
OLS-4	Unrestricted	0.460	0.557	0.287	1/10; 2/10
OLS-4	Restricted	0.420	0.776	0.220	1/10; 2/10
Ridge	Unrestricted	0.280	0.894	0.167	3/10; 5/10
Ridge	Restricted	0.300	1.019	0.157	4/10; 5/10
Elastic Net	Unrestricted	0.700	0.765	0.229	1/10; 2/10
Elastic Net	Restricted	0.520	0.898	0.224	2/10; 3/10

Note. The relative adjusted difference is normalized by the unrestricted OLS model, whose average adjusted difference is 0.000050. Lower p -values indicate stronger evidence that the model improves forecast accuracy relative to the historical mean benchmark. The median p -value provides a compact summary that is less influenced by unusually large or small industry-specific p -values. The best-performing model is highlighted in bold.

The Clark–West test compares a forecasting model against a nested benchmark, which in this case is the historical mean. It tests whether the model provides statistically better out-of-sample forecasts after adjusting for the estimation error faced by larger models. The null hypothesis is that the model does not improve forecast accuracy relative to the historical mean, while the alternative is that it does. Table 2 indicates that Ridge provides the strongest evidence of improvement. The restricted Ridge model has the highest average Clark–West statistic and the lowest median p -value, with significant results for 4 industries at the 5% level and 5 industries at the 10% level. Elastic Net and OLS-based models show weaker evidence, although the restricted specifications generally perform slightly better than the unrestricted ones. Overall, the test suggests that predictive gains are present, especially for Ridge, but they are not uniform across all industries.

Table 3: Selected Diebold–Mariano Forecast Comparisons

Comparison	Avg. DM	Median p	Significant 5%;10%
<i>Restricted vs unrestricted forecasts</i>			
OLS Restricted vs OLS	2.047	0.030	7/10; 7/10
OLS-4 Restricted vs OLS-4	1.513	0.157	1/10; 2/10
Ridge Restricted vs Ridge	1.199	0.254	0/10; 1/10
Elastic Net Restricted vs Elastic Net	1.520	0.137	2/10; 3/10
<i>Selected model comparisons</i>			
OLS-4 Restricted vs OLS Restricted	1.067	0.271	0/10; 1/10
OLS-4 Restricted vs Elastic Net Restricted	0.498	0.466	0/10; 0/10
Ridge Restricted vs OLS Restricted	1.610	0.104	4/10; 5/10
Ridge Restricted vs OLS-4 Restricted	0.788	0.391	0/10; 0/10
Ridge Restricted vs Elastic Net Restricted	1.089	0.245	1/10; 2/10
Random Forest vs Historical Mean	-0.527	0.623	1/10; 1/10
Random Forest vs OLS-4 Restricted	-0.204	0.534	0/10; 0/10
Random Forest vs Ridge Restricted	-0.857	0.336	1/10; 1/10
Random Forest vs Elastic Net Restricted	0.270	0.622	0/10; 0/10

Note. Positive average DM statistics indicate lower average forecast loss for the first model in the comparison, while negative values indicate lower average forecast loss for the second model.

The Diebold-Mariano results show that the non-negativity restriction improves forecast accuracy most clearly for the full OLS model, where the restricted specification significantly outperforms the unrestricted version in seven out of ten industries. The gains from restrictions are weaker for OLS-4, Ridge, and Elastic Net, suggesting that forecast truncation adds less once the model is already simplified or regularized. Among the selected restricted models, Ridge performs better than restricted OLS in several industries, but it does not significantly dominate restricted OLS-4 or restricted Elastic Net overall. Random

Forest does not statistically outperform the historical mean or the strongest restricted linear benchmarks. Its negative average DM statistics against the historical mean, restricted OLS-4, and restricted Ridge indicate weaker average forecast performance, although the differences are generally not statistically significant across most industries. These results suggest that the advantage of model flexibility is limited in this monthly industry-portfolio setting. This differs from the broader machine-learning evidence in [Gu et al. \(2020\)](#), but is consistent with the view that complex models do not automatically improve forecasts when return signals are weak and noisy.

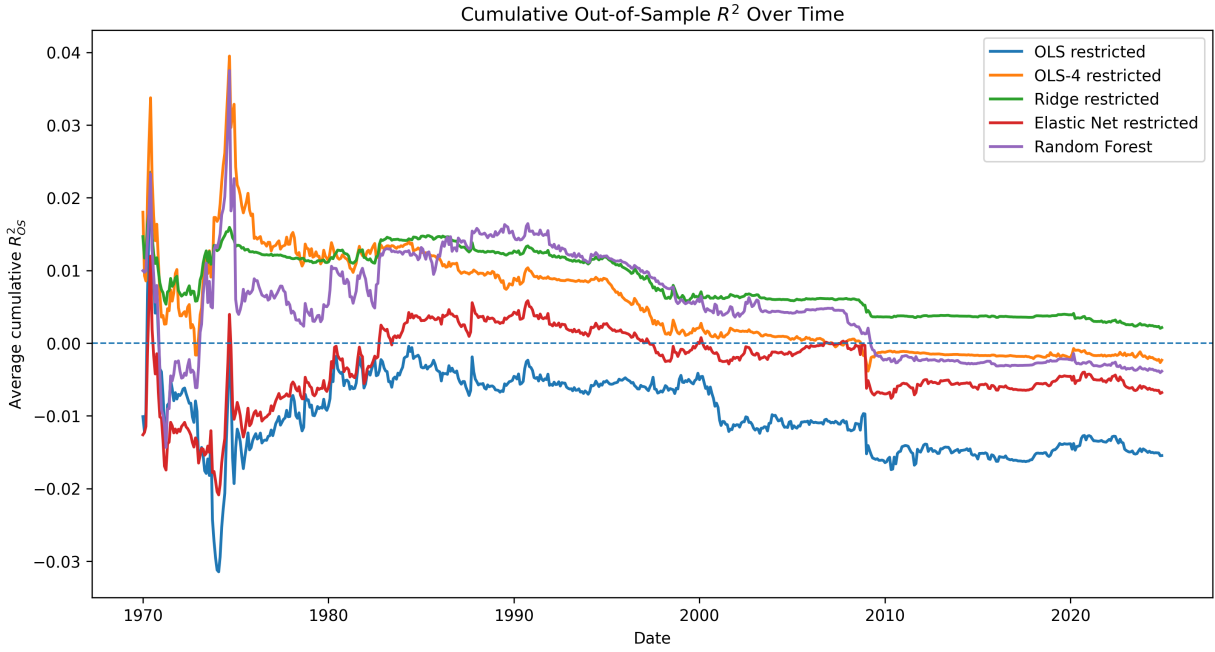


Figure 1: Forecast Performance Over Time

The figure 1 plots the cumulative out-of-sample R^2 over time, averaged across the 10 industry portfolios. The figure starts in 1970 because cumulative R^2_{OOS} is very sensitive to individual forecast errors at the beginning of the out-of-sample period, when cumulative error sums are still small.

The results show that forecast performance is time-varying and generally close to zero. Ridge is the most consistently positive model for much of the sample, while restricted OLS remains mostly negative. Random Forest performs relatively well during parts of the 1980s and early 1990s, but its advantage declines later in the sample.

The largest movements occur in the early 1970s and around 1973–1975. The early movements mainly reflect cumulative R_{OOS}^2 sensitivity when only a limited number of observations have accumulated. The movements around 1973–1975 coincide with the first oil shock and the associated recession, suggesting that forecast performance is affected by volatile macroeconomic conditions. The asset-level results show that several of the largest changes around January 1975 are concentrated in industries such as Shops (wholesale and retail), NoDur (consumer non-durables), Durbl (consumer durables), and HiTec (business equipment). This suggests that the aggregate movement in cumulative R_{OOS}^2 is partly driven by sector-specific forecast errors during the recession and oil-shock period, rather than by all industries moving equally.

Overall, the forecasting results answer the first part of the research question cautiously. Machine learning does not clearly improve monthly industry-return forecasts in this setting, while regularization provides more reliable gains. Non-negativity restrictions generally improve linear forecasts, but the improvements are modest and not always statistically significant. Ridge is the most consistent forecasting model across MSPE, R_{OOS}^2 , Clark-West tests. Random Forest, despite its flexibility, does not outperform the historical mean or the strongest linear benchmarks in forecast accuracy, suggesting that model complexity alone is not sufficient to overcome the noise in monthly returns.

4.2 Portfolio Evaluation

The economic importance of return predictability is not captured by statistical accuracy alone. [Campbell and Thompson \(2008\)](#) show that even modest levels of return predictability may generate meaningful benefits for a mean–variance investor. They argue that predictive R^2 should be evaluated relative to the squared Sharpe ratio of the risky asset, since small improvements in forecast accuracy can translate into better risk-adjusted investment performance. In their framework, an investor who uses predictive information to time the market can achieve an improved Sharpe ratio:

$$SR^* = \sqrt{\frac{SR^2 + R_{OOS}^2}{1 - R_{OOS}^2}}. \quad (6)$$

Because the squared Sharpe ratio of stock returns is typically small, even a monthly

R_{OOS}^2 of only 0.5% may be economically meaningful. Nevertheless, the relationship is not automatic: forecast improvements may fail to generate economic value if they result in unstable portfolio weights, excessive turnover, or high transaction costs. Recent research also suggests that profitable market-timing strategies can sometimes emerge even when R_{OOS}^2 is weak or negative (Kelly et al., 2024). For this reason, the empirical analysis evaluates models not only through forecast accuracy, but also through portfolio-based measures such as the Sharpe ratio, certainty-equivalent return, and turnover.

The analysis focuses on the restricted versions of the linear models, since the forecasting results show that restrictions generally improve or stabilize performance. These strategies are compared with portfolios based on the historical mean, Random Forest forecasts, and the (1/N) benchmark. To account for the instability of unconstrained mean-variance weights, the results are reported across several portfolio specifications, including long-only portfolios, capped-weight portfolios with maximum industry weights of 20%, 30%, and 50%, and combinations with the 1/N benchmark using different shrinkage intensities ($\lambda = 0.25, 0.50, \text{ and } 0.75$). This allows the analysis to assess whether forecasts create economic value once portfolio concentration and trading intensity are controlled.

Table 4: Portfolio Performance: Unrestricted and Long-Only Strategies

Model	Portfolio constraint	Ann. Return	Ann. Volatility	Sharpe	CER
1/N	Benchmark	0.076	0.151	0.506	0.042
Historical mean	Unrestricted	0.065	0.144	0.451	0.034
Historical mean	Long-only	0.062	0.146	0.423	0.030
OLS	Unrestricted	0.262	2.966	0.088	-12.930
OLS	Long-only	0.073	0.181	0.406	0.024
OLS-4	Unrestricted	0.126	1.530	0.082	-3.386
OLS-4	Long-only	0.072	0.192	0.373	0.016
Ridge	Unrestricted	0.076	0.243	0.311	-0.013
Ridge	Long-only	0.071	0.163	0.439	0.032
Elastic Net	Unrestricted	-0.743	3.655	-0.203	-20.784
Elastic Net	Long-only	0.079	0.184	0.430	0.028
Random Forest	Unrestricted	0.976	5.234	0.187	-40.109
Random Forest	Long-only	0.074	0.180	0.411	0.025

Note. Bold rows indicate the best overall strategy and the strongest unrestricted and long-only forecast-based strategies by risk-adjusted performance. All strategies are evaluated over 696 monthly observations.

Table 4 shows that unrestricted portfolios are highly unstable, especially for Random Forest, OLS, and Elastic Net. Although some unrestricted strategies generate high average returns, these are accompanied by extremely large volatility and strongly negative certainty-equivalent returns. This suggests that unconstrained mean–variance optimization produces unrealistic and overly aggressive portfolio weights.

Imposing long-only constraints substantially stabilizes the portfolios. Volatility and downside outcomes fall sharply across all forecast-based models. However, even after this improvement, none of the long-only forecast-based portfolios outperform the 1/N benchmark in terms of Sharpe ratio or certainty-equivalent return. This indicates that basic non-negativity constraints are useful, but not sufficient to make the model-based portfolios clearly superior to equal weighting.

Table 5: Best-Performing Portfolio Strategies by Forecasting Model

Model	Portfolio restriction	Ann. Return	Ann. Volatility	Sharpe	CER
$1/N$	Benchmark	0.076	0.151	0.506	0.042
Historical mean	Long-only + $\lambda(= 0.25)$	0.073	0.148	0.491	0.040
Historical mean	Cap (20%)	0.067	0.143	0.469	0.036
Historical mean	Cap (20%) + $\lambda(= 0.25)$	0.074	0.148	0.499	0.041
OLS	Long-only + $\lambda(= 0.25)$	0.076	0.151	0.501	0.041
OLS	Cap (30%)	0.083	0.152	0.544	0.048
OLS	Cap (30%) + $\lambda(= 0.75)$	0.081	0.150	0.542	0.048
OLS-4	Long-only + $\lambda(= 0.25)$	0.075	0.155	0.484	0.039
OLS-4	Cap (20%)	0.079	0.154	0.516	0.044
OLS-4	Cap (20%) + $\lambda(= 0.75)$	0.079	0.152	0.516	0.044
Ridge	Long-only + $\lambda(= 0.25)$	0.075	0.150	0.501	0.041
Ridge	Cap (20%)	0.076	0.149	0.512	0.043
Ridge	Cap (20%) + $\lambda(= 0.75)$	0.076	0.148	0.513	0.043
Elastic Net	Long-only + $\lambda(= 0.25)$	0.077	0.152	0.507	0.042
Elastic Net	Cap (30%)	0.083	0.154	0.536	0.047
Elastic Net	Cap (30%) + $\lambda(= 0.75)$	0.081	0.151	0.535	0.047
Random Forest	Long-only + $\lambda(= 0.25)$	0.076	0.152	0.497	0.041
Random Forest	Cap (20%)	0.081	0.149	0.544	0.048
Random Forest	Cap (20%) + $\lambda(= 0.75)$	0.080	0.148	0.538	0.047

Note. The table reports the best-performing specification within each restriction type for each forecasting model. The shrinkage parameter λ denotes the weight placed on the model-based portfolio, with the remaining weight allocated to the $1/N$ benchmark. Bold rows indicate the five strongest forecast-based portfolios by risk-adjusted performance.

Table 5 summarizes the main constrained portfolio results. To keep the comparison concise, the table reports only the best-performing specification within each restriction type for every model, rather than all possible cap and shrinkage combinations. The results show that simple long-only shrinkage portfolios remain very close to the $1/N$ benchmark

and provide only limited improvements. In contrast, capped portfolios perform considerably better. Several forecast-based strategies outperform $1/N$ in terms of Sharpe ratio and certainty-equivalent return once maximum industry weights are imposed. The strongest results are obtained by the 30% capped OLS portfolio and the 20% capped Random Forest portfolio, while Elastic Net also performs well under a 30% cap. The strong performance of capped portfolios in this analysis supports the more recent evidence of [Babiarz and Baruník \(2026\)](#), who emphasize that weight restrictions can stabilize forecast-based portfolios and improve economic performance. Adding shrinkage toward $1/N$ to the capped portfolios generally improves stability, but it does not clearly outperform the best capped-only specifications.

The portfolio results show that weak statistical forecast performance does not necessarily rule out economic value consistent with [Kelly et al. \(2024\)](#). Several forecast-based strategies perform better and manage to outperform $1/N$ strategy once the allocation problem is controlled through weight caps. However, these gains do not appear in unrestricted portfolios, where estimation error leads to extreme portfolio weights, nor in simple long-only portfolios, which remain close to the $1/N$ benchmark. Overall, the results suggest that return forecasts become economically useful only when combined with portfolio constraints that limit concentration and mitigate the impact of noisy expected-return estimates.

4.3 Turnover and Costs

Table 6: Net Returns After Transaction Costs

Strategy	Turnover	Gross Ret.	Net Ret. 10	Net Ret. 25	Net Ret. 50
$1/N$	0.012	0.076	0.076	0.076	0.076
HM cap 20% + $\lambda = 0.25$	0.012	0.074	0.074	0.074	0.073
OLS cap 30%	0.252	0.083	0.080	0.075	0.068
OLS-4 cap 20%	0.063	0.079	0.079	0.077	0.076
Ridge cap 20% + $\lambda = 0.75$	0.105	0.076	0.075	0.073	0.070
EN cap 30%	0.267	0.083	0.079	0.075	0.067
RF cap 20%	0.227	0.081	0.078	0.074	0.067

Note. The table reports average monthly turnover, gross annualized excess return, and net annualized excess returns after transaction costs of 10, 25, and 50 basis points. HM denotes historical mean, EN denotes Elastic Net, and RF denotes Random Forest. Bold entries highlight the lowest-turnover benchmark and the strongest forecast-based strategies based on net Sharpe ratios and net CER. All strategies are evaluated over 696 monthly observations.

Transaction costs reduce the advantage of the forecast-based portfolios. Before costs, the strongest strategies are OLS cap 30%, Elastic Net cap 30%, and Random Forest cap 20%, all of which have higher gross annual returns than the $1/N$ benchmark. However, these strategies also have much higher turnover. For example, average monthly turnover is only 0.012 for $1/N$, compared with 0.252 for OLS cap 30%, 0.267 for Elastic Net cap 30%, and 0.227 for Random Forest cap 20%.

At low transaction costs, forecast-based portfolios still remain attractive. OLS cap 30% has the highest net annual return at 10 basis points, while Random Forest and OLS also maintain stronger net Sharpe ratios than $1/N$ at this cost level. However, under the conservative 50 basis point cost assumption, the $1/N$ portfolio becomes the most robust strategy. It has the highest net annual return, the highest net Sharpe ratio, and the highest net CER. This shows that forecast-based portfolios can create economic value, but their advantage is sensitive to trading costs because they require substantially more rebalancing than the equal-weight benchmark. As a result, the economic value of return forecasts depends not only on predictive information, but also on whether the resulting portfolio

can be implemented without excessive turnover. This is consistent with DeMiguel et al. (2009) and Tu and Zhou (2011), who show that implementation frictions and estimation error help explain why the $1/N$ benchmark remains difficult to beat.

4.4 Rolling-Window Robustness

As mentioned in Methods section 3.6 we completed robustness check based on 20 year rolling-window, forecasting results for same months as using the expanding window.

Table 7: Rolling-Window Forecasting Performance Across Industry Portfolios

Model	Restriction	Rel. MSPE	Average R^2_{OOS}	Positive R^2_{OOS}
OLS	No	1.000	-0.1064	0/10
OLS	Yes	0.929	-0.0290	1/10
OLS-4	No	0.920	-0.0200	0/10
OLS-4	Yes	0.896	0.0075	8/10
Ridge	No	0.896	0.0065	8/10
Ridge	Yes	0.895	0.0081	8/10
Elastic Net	No	0.904	-0.0013	5/10
Elastic Net	Yes	0.902	0.0006	6/10
Random Forest	No	0.911	-0.0087	1/10

Note. Relative MSPE is normalized by the unrestricted OLS model, whose average MSPE is 0.003350, so values below 1 indicate lower average forecast errors than unrestricted OLS. The restriction refers to truncating negative predicted excess-return forecasts at zero. The best-performing model is highlighted in bold.

Table 7 reports the forecast results under rolling-window specification. The results are broadly consistent with the main expanding-window analysis. Ridge remains the strongest forecasting model, with the restricted version producing the highest average R^2_{OOS} and positive R^2_{OOS} for 8 out of 10 industries. OLS-4 restricted also performs well, achieving positive average R^2_{OOS} and positive predictive performance in 8 industries.

The results again show that forecast restrictions are useful. OLS, OLS-4, Ridge, and

Elastic Net all improve after imposing the non-negativity restriction, although the gains are small. In contrast, Random Forest performs poorly in terms of forecast accuracy under the rolling-window specification, with negative average R^2_{OOS} and only one industry showing positive R^2_{OOS} .

Compared with the main expanding-window results, the rolling-window results preserve the main forecasting conclusion: predictive gains are modest and strongest for Ridge, while restrictions generally improve performance. However, Random Forest becomes weaker in forecast accuracy, suggesting that it is more sensitive to the shorter rolling estimation window.

Table 8: Diebold-Mariano Tests: Rolling vs. Expanding Forecasts

Model	Avg. DM stat.	Median p	Sig. 5%	Sig. 10%	Rolling better
OLS	-1.363	0.155	3	4	1
OLS restricted	-0.723	0.584	2	3	3
OLS-4	-0.818	0.448	0	1	0
OLS-4 restricted	0.272	0.598	0	0	7
Ridge	-0.097	0.609	1	1	6
Ridge restricted	0.174	0.518	2	2	7
Elastic Net	1.039	0.328	2	2	10
Elastic Net restricted	0.257	0.535	0	0	7
Random Forest	-1.045	0.249	0	1	1

Note. The table summarizes Diebold–Mariano tests comparing rolling-window and expanding-window forecasts for the same model. Positive average DM statistics indicate lower squared forecast errors for the rolling-window specification, while negative values indicate lower squared errors for the expanding-window specification. “Rolling better” reports the number of industries for which the rolling-window forecast has lower squared forecast error by sign.

Table 8 compares rolling-window and expanding-window forecasts using Diebold–Mariano tests. The results suggest that the choice of estimation window does not lead to a clear, systematic improvement across all models. For OLS and Random Forest, the average DM statistics are negative and expanding-window forecasts have lower squared errors for most industries. This suggests that these models benefit more from the larger expanding training

sample.

In contrast, restricted OLS-4, restricted Ridge, and restricted Elastic Net have positive average DM statistics, and rolling-window forecasts perform better by sign for 7 out of 10 industries in each case. Elastic Net also shows the strongest rolling-window improvement by sign, with rolling forecasts outperforming expanding forecasts in all 10 industries for the unrestricted version. However, the median p -values are generally high, and only a small number of industry-level differences are statistically significant.

Overall, the results indicate that rolling-window estimation does not significantly dominate expanding-window estimation. Instead, the two approaches produce broadly similar forecast accuracy. This supports the robustness of the main findings, because the forecasting results are not driven entirely by the choice of expanding-window estimation.

Table 9: Rolling-Window Portfolio Performance

Strategy	Ann. Return	Ann. Volatility	Sharpe	CER
$1/N$	0.076	0.151	0.506	0.042
Historical mean cap 20% + $\lambda = 0.25$	0.074	0.147	0.503	0.041
OLS cap 30%	0.093	0.154	0.603	0.057
OLS-4 cap 20%	0.086	0.150	0.570	0.052
Ridge cap 20% + $\lambda = 0.75$	0.077	0.144	0.538	0.046
Elastic Net cap 30%	0.063	0.143	0.438	0.032
Random Forest cap 20%	0.082	0.152	0.538	0.047

Note. The table reports portfolio performance using 20-year rolling-window forecasts over the same 696-month evaluation sample as the main analysis. The portfolio specifications are selected from the best-performing strategies in the main analysis and then re-estimated using rolling-window forecasts. Bold indicates the best-performing strategy by Sharpe ratio and CER.

Table 9 reports the portfolio robustness results using 20-year rolling-window forecasts. The purpose is not to select new optimal portfolios, but to test whether the best-performing specifications from the main analysis remain economically useful when the estimation window is changed.

The results support the main portfolio conclusion. Several forecast-based portfolios

continue to outperform the $1/N$ benchmark in risk-adjusted terms. The strongest performance is obtained by the OLS cap 30% portfolio, which achieves an annualized return of 9.3%, a Sharpe ratio of 0.603, and a CER of 0.057. OLS-4 cap 20% also performs well, with a Sharpe ratio of 0.570 and a CER of 0.052. Random Forest and Ridge remain competitive, both achieving Sharpe ratios above the $1/N$ benchmark.

Compared with the main expanding-window analysis, the rolling-window results strengthen the performance of OLS and OLS-4 portfolios, while Random Forest remains positive but is no longer the leading strategy. Elastic Net performs worse under rolling estimation, suggesting that its strong main-analysis portfolio result is less robust to the shorter estimation window. Overall, the results indicate that the economic value of forecast-based portfolios is not driven solely by the expanding-window design.

5 Discussion

5.1 Discussion of Hypotheses

The results provide mixed support for the hypotheses. H1 is supported: unrestricted OLS predictive regressions do not outperform the historical mean benchmark in monthly out-of-sample expected-return forecasting. The unrestricted OLS models have negative average out-of-sample R^2 values, confirming that simple predictive regressions are unreliable in this setting. This is consistent with [Welch and Goyal \(2008\)](#) and [Gu et al. \(2020\)](#), who show that unrestricted OLS forecasts often fail out of sample because of estimation error and overfitting.

H2 is partially supported. Restricted predictive regressions generally perform better than unrestricted predictive regressions. This is clearest for OLS-4, where the restricted version achieves positive average R_{OS}^2 and increases the number of industries with positive R_{OS}^2 . However, the improvement over the historical mean benchmark remains small and is not consistently statistically significant. The result therefore moves in the same direction as [Campbell and Thompson \(2008\)](#), [Li and Tsiakas \(2017\)](#), and [?](#), who show that economic restrictions can stabilize forecasts and reveal weak predictive signals.

H3 is partially supported. Regularized linear models outperform unrestricted OLS and

the historical mean more clearly than the simple predictive regressions. Ridge provides the strongest forecasting results among all models, both in the main expanding-window analysis and in the rolling-window robustness check. Elastic Net, however, performs much weaker, with average R_{OS}^2 close to zero. Therefore, the hypothesis is supported mainly by Ridge, not by both regularized models equally. This is consistent with the broader evidence that regularization can reduce instability in high-dimensional regressions (Gu et al., 2020; Zou and Hastie, 2005).

H4 is not supported. Random Forest forecasts do not outperform the historical mean benchmark, restricted predictive regressions, or regularized linear models in monthly out-of-sample forecasting. Its average R_{OS}^2 is weak in the main analysis and remains negative in the rolling-window robustness check. This contrasts with the stronger machine learning evidence in Gu et al. (2020), and suggests that model flexibility alone is not sufficient to improve monthly industry return forecasts.

H5 is partially supported. Portfolio strategies based on regularized linear forecasts perform well once portfolio constraints are imposed. Elastic Net cap 30% achieves a Sharpe ratio of 0.536 and a CER of 0.047, both above the $1/N$ benchmark and above the historical mean portfolio. Ridge-based portfolios are more stable and perform better than the historical mean, but they are less dominant than Ridge is in forecast accuracy. Overall, the result supports the view that regularized forecasts can add portfolio value, but mainly when combined with weight constraints.

H6 is also partially supported. Random Forest-based portfolio strategies perform strongly under weight caps, especially the 20% capped Random Forest portfolio, which has one of the highest Sharpe ratios in the main analysis. However, Random Forest does not clearly dominate the competing forecast-based portfolios. OLS cap 30% achieves a slightly higher CER in the main analysis and performs better under rolling-window estimation. Therefore, Random Forest creates economic value in constrained portfolios, but it is not consistently superior to restricted predictive regressions or regularized linear models. This is consistent with ?, who show that weak R_{OS}^2 does not necessarily rule out economic value.

H7 is partially supported. Before transaction costs, the best forecast-based portfolios outperform $1/N$ in gross performance. However, after accounting for turnover and trans-

action costs, the advantage becomes weaker. At low transaction costs, OLS cap 30% and Random Forest cap 20% remain attractive. Under the conservative 50 basis point cost assumption, however, $1/N$ becomes the most robust strategy because it has much lower turnover. This confirms that forecast-based portfolios can be economically meaningful, but their value is sensitive to trading frictions. The result is consistent with the literature showing that estimation error and trading frictions make the naive $1/N$ portfolio difficult to beat (DeMiguel et al., 2009; Tu and Zhou, 2011).

5.2 Role of Portfolio Constraints

Portfolio constraints are essential in the results. Unrestricted mean–variance portfolios are unstable, especially for Random Forest, OLS, and Elastic Net. These strategies sometimes generate high average returns, but with extreme volatility and strongly negative CER values. This is consistent with the classic concern that mean–variance optimization is highly sensitive to estimation error in expected returns (Markowitz, 1952; DeMiguel et al., 2009). Long-only constraints improve stability, but they are not enough to beat the $1/N$ benchmark. This finding is in line with DeMiguel et al. (2009), who show that optimized portfolios often struggle to outperform naive diversification out of sample, and with Tu and Zhou (2011), who argue that combining naive and optimized portfolio rules can reduce estimation risk. The most important improvement comes from maximum weight caps. Capped portfolios prevent excessive concentration and allow forecasts to influence the portfolio without creating extreme positions. The best results are obtained by OLS cap 30%, Random Forest cap 20%, and Elastic Net cap 30%. This supports the conclusion that forecasts become economically useful mainly when combined with diversification constraints.

5.3 Transaction Costs and Practical Relevance

Transaction costs reduce the advantage of forecast-based portfolios. Before costs, OLS cap 30%, Elastic Net cap 30%, and Random Forest cap 20% all generate higher gross returns than $1/N$. However, these strategies also have much higher turnover. Average monthly turnover is only 0.012 for $1/N$, compared with 0.252 for OLS cap 30%, 0.267 for Elastic

Net cap 30%, and 0.227 for Random Forest cap 20%. This is consistent with [DeMiguel et al. \(2009\)](#), who show that optimized portfolios often have much higher turnover than the naive $1/N$ rule. It is also in line with [Gu et al. \(2020\)](#), who find that machine learning portfolios can generate attractive performance but often require substantial trading.

At low transaction costs, forecast-based portfolios remain attractive. OLS cap 30% still has the highest net annual return at 10 basis points. However, under the conservative 50 basis point assumption, the $1/N$ benchmark becomes the most robust strategy. This result is consistent with [DeMiguel et al. \(2009\)](#), who use transaction costs to show that the gains from optimized portfolios can be offset by turnover. It also supports the broader point that portfolio gains from return predictability are economically meaningful only if they survive realistic trading frictions ([Gu et al., 2020](#); [Tsiakas et al., 2020](#)). Overall, the results show that forecast-based portfolios can create economic value, but their practical advantage depends strongly on trading costs.

5.4 Robustness

The rolling-window robustness results show that the main conclusions are not driven only by the expanding-window design. Forecast gains remain modest, and Ridge remains the strongest forecasting model. Random Forest performs worse in terms of rolling-window forecast accuracy, which suggests that it is sensitive to the shorter estimation window.

The portfolio robustness results are stronger. Using the same best-performing portfolio specifications from the main analysis, capped forecast-based portfolios continue to perform well under rolling-window forecasts. The OLS cap 30% portfolio performs especially strongly, with a Sharpe ratio of 0.603 and a CER of 0.057. This supports the main conclusion that forecast-based portfolios can have economic value, but mainly when combined with weight constraints.

6 Conclusion

This thesis examined whether machine learning and regularized linear models can outperform traditional benchmarks in monthly out-of-sample industry return forecasting, and

whether these forecasts can generate economically valuable portfolio strategies. The analysis therefore connects two questions: whether models improve forecast accuracy, and whether any forecasting gains survive the portfolio construction process.

The forecasting evidence is mixed. Unrestricted OLS performs poorly, while non-negativity restrictions improve the linear models, especially OLS-4. Among all models, Ridge provides the strongest and most robust forecasting performance. Random Forest, despite being the main nonlinear machine learning model, does not clearly outperform the historical mean, restricted predictive regressions, or regularized linear models in MSPE terms. This suggests that, in this setting, regularization is more useful than nonlinear model flexibility for improving monthly return forecasts.

The portfolio results show that weak forecast accuracy does not necessarily imply zero economic value. Some models with weak average R_{OOS}^2 , especially Random Forest and Elastic Net, perform well once their forecasts are used in constrained portfolio allocation. However, this value appears only when the optimization problem is controlled. Unrestricted mean-variance portfolios are unstable, and long-only constraints alone are not enough to beat the $1/N$ benchmark. The strongest results come from capped portfolios, especially OLS cap 30%, Random Forest cap 20%, and Elastic Net cap 30%.

Taken together, the main finding is not that machine learning clearly dominates traditional forecasting methods. Instead, the evidence shows that forecast-based allocation can add value when noisy forecasts are combined with restrictions, regularization, and portfolio constraints. Model complexity alone is not sufficient. Simpler models, especially restricted OLS and Ridge, are often competitive with or better than Random Forest.

Transaction costs are an important qualification. Forecast-based portfolios trade more than $1/N$, so their advantage is sensitive to implementation costs. At low transaction costs, some model-based strategies remain attractive. Under the conservative 50 basis point assumption, however, $1/N$ becomes the most robust strategy. The rolling-window robustness analysis leads to the same broader conclusion: capped forecast-based portfolios remain useful, but no single forecasting method dominates across all settings.

Overall, the evidence does not support the view that greater model complexity leads to better monthly return forecasts. Random Forest fails to consistently outperform sim-

pler alternatives, while Ridge delivers the most reliable forecasting performance across the evaluation metrics. Nevertheless, the portfolio analysis demonstrates that even modest forecasting signals can be valuable when they are incorporated into a disciplined investment process with appropriate constraints.

Future research could extend this analysis in several ways. First, portfolio construction could be improved by using alternative covariance estimators or more advanced shrinkage methods. Second, the analysis could be repeated with individual stocks or international portfolios to test whether the results generalize beyond U.S. industry portfolios. This extension could also include a broader predictor set, such as firm-level characteristics related to size, value, and momentum, to examine whether machine learning methods perform better when richer cross-sectional information is available. Third, future work could incorporate more realistic transaction-cost estimates that vary across time, market conditions, and asset liquidity. Finally, additional machine learning models could be evaluated, but the results of this thesis suggest that model complexity alone is unlikely to be sufficient unless it is combined with careful portfolio constraints.

7 References

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8 Appendix

Net Sharpe ratios and Net CER from table 6:

Table 10: Transaction Cost Performance: Sharpe Ratios and CER

Strategy	Gross Sharpe	Gross CER	Sharpe 10	CER 10	Sharpe 50	CER 50
$1/N$	0.506	0.042	0.505	0.042	0.501	0.041
HM cap 20% + $\lambda = 0.25$	0.499	0.041	0.498	0.041	0.494	0.040
OLS cap 30%	0.544	0.048	0.524	0.045	0.444	0.033
OLS-4 cap 20%	0.516	0.044	0.511	0.043	0.491	0.040
Ridge cap 20% + $\lambda = 0.75$	0.513	0.043	0.504	0.042	0.470	0.037
EN cap 30%	0.536	0.047	0.515	0.044	0.431	0.031
RF cap 20%	0.544	0.048	0.526	0.045	0.452	0.034

Note. The table reports gross and net Sharpe ratios and certainty-equivalent returns after transaction costs of 10 and 50 basis points. HM denotes historical mean, EN denotes Elastic Net, and RF denotes Random Forest. All strategies are evaluated over 696 monthly observations.